Climate Finance

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Plan

I

- Using satellite data to empirically investigate questions related to preservation of tropical forests
  1. Effect of national policies on deforestion
  2. Effect of law enforcement against illegal deforestation in the Amazon.
  3. Understanding spacial amplification of degradation in the forest.
- Discuss empirical strategies that may be useful in other contexts.

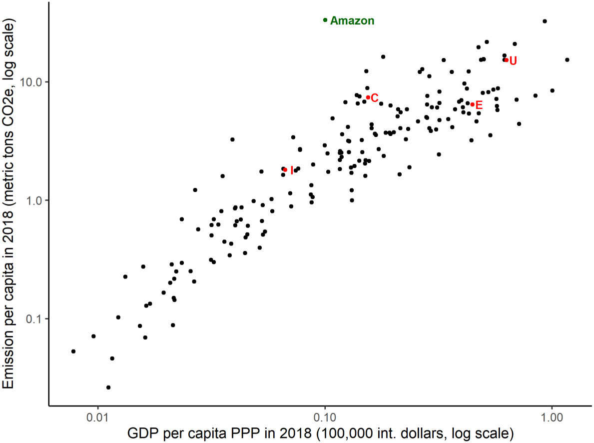
Tropical Forest

- Tropical forests contain very large amounts of captured carbon.
  - Saatchi et al. [2011] estimates that 247 Gt C, with 193 Gt C stored above ground and 54 Gt C stored below ground in roots.
  - 1 Gigaton=1 billion tonnes.
- Total historical emissions of US $\sim$ 450 GT of CO_2.
  - Ratio of the mass of carbon to that of carbon dioxide = 3.667

Brazilian Amazon

- Malhi et al. [2006] estimates that the Amazon forest contains 123 ± 31 Gt of captured carbon.
- 4.2 million km^2 or 60% of Amazon is in Brazil.
- An area the size of Texas has been deforested in the Brazilian Amazon.
  - Logging
  - In the 21st century, mostly for cattle.
    - Land speculation

Emissions curve

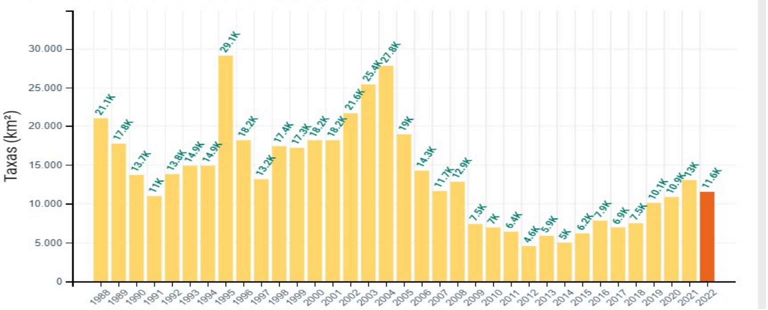


Deforestation in the Brazilian amazon

I

- Bulk of deforestation is illegal (Assunção et al. [2017])
- Law enforcement has varied over time as a result of politics and economic conditions.
- Variations in law enforcement led to variations in deforestation rates.
- Burgess et al. [2019] document two reversals between in 2000-2018

Deforestation Brazilian Legal Amazon in Km^2



- `https://web.archive.org/web/20230110063026/http://terrabrasilis.dpi.inpe.br/app/dashboard/deforestation/biomes/legal_amazon/rates`

Brazilian Amazon two reversals

- Burgess et al. [2019]. Example on how to use large satellite data sets to study effects of national policies in tropical forests.
- Uses annual 30mx30m resolution Landsat 7 data to measure deforestation over time and space from 2000 to 2018.
- In 2000, Brazilian plots 37% more likely to have been deforested than similar lands located just a few kilometers away across the border.
- From 2001 to 2005, the annual deforestation rate was more than three times higher on the Brazilian side of the border than in neighbors.

Brazilian Amazon two reversals II

- Differences similar across the borders with both Bolivia and Peru indicating that differences are due to policies in Brazil, rather than in countries across the border.
- In late 2004, Brazil strengthens legal penalties associated with illegal deforestation.
- In 2006 Brazil starts using satellite-based detection of deforestation and police and army enforcement operations targeted to detected illegal deforestation.
- Discontinuity in deforestation rates disappears precipitously in 2006 and is not present until 2013.

Brazilian Amazon two reversals

III

- In late 2012, Brazil passes "New Forest Code" that, in practice, forgave 90% of pre-2008 deforestation.
- In 2014 Brazil implemented regulatory changes that lowered commitment to environmental preservation.
- Budget of the Brazilian Environmental Agency (IBAMA) reduced by 43% between 2013 and 2016.
- Starting in 2014, deforestation rates in Brazil started to return to near pre-2004 levels.
- Situation got worse under Bolsonaro (2019-2022), see above.

Empirical strategy

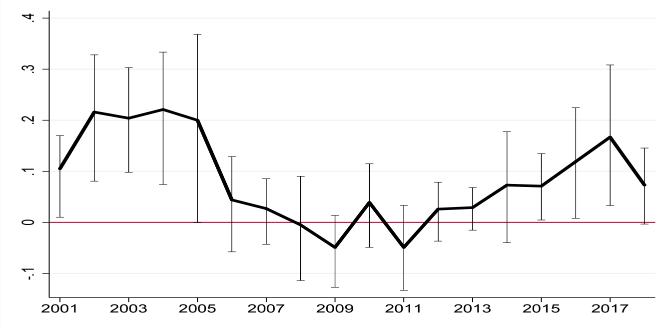
I

- Basic regressions:

$$\begin{aligned} Y_i &= \alpha + \gamma B_i + f(DistBorder_i) + \delta X_i + \epsilon_i \\ f(DistBorder_i) &= B_i * f^B(DistBorder_i) + (1 - B_i) f^{NB}(DistBorder_i) \end{aligned}$$

  Y_i is outcome, (% forest lost as of 2000 or % loss in year), B_i is Brazil dummy and X_i are additional controls (distance from water, slope,...) Not treated as panel.
- γ coefficient of interest in each regression.
- Choose bandwith "optimally" which is 25km from border.
- Construct sites of 120mx120m (16 original pixels).
- Left with 31,071,838 observations within 1,094 clusters of 25 km x 25 km (use clusters to calculate standard errors)

Estimates of differences in percentage points of % loss (γ) and SE



(a) Overall Effects (OLS model)

Estimates of differences in percentage points of % loss (γ) and SE

II

- Do not believe results that model deforestation as choices of land-grabbers in a constant environment.

Law enforcement and illegal deforestation

I

- Endogeneity
  - Presence of law enforcement negatively impacts illegal forest clearing by inhibiting potential offenders.
  - Law enforcers are allocated partially based on the observation of clearings.
  - Only observe "'equilibrium" outcomes
- Coefficient of law enforcement on a OLS regression of change in forest cover on law enforcement plus controls cannot be interpreted as casual impact of law enforcement on forest clearing.
- Instruments.

Law enforcement and illegal deforestation II

- Assunção et al. [2017] uses average daily cloud covers that inhibit satellite detection in particular areas as instrument for law enforcement.
    - Cloud cover is not affected by current deforestation.

Empirical strategy

I

- Measure law enforcement by fines applied.
- Panel of observations over municipalities i and time t (between 2006 and 2016.)
- Deforestation is measured by PRODES that uses a better (and more expensive) satellite system, and chooses the best pictures from the Amazon dry season for each area.
- OLS regression

$$Deforest_{i,t} = \tilde{\beta} LE_{i,t-1} + \sum_k \left[\tilde{\gamma}_k Control_{i,t,k}\right] + \tilde{\alpha}_i + \tilde{\phi}_t + \tilde{\epsilon}_{i,t} \tag{1}$$

Empirical strategy

II

- Controls include, agricultural prices at municipality, precipitation and temperature at municipality, and PRODES satellite blocked areas.
- First-stage regression

$$LE_{i,t} = \beta Cloudc_{i,t} + \sum_{k} [\gamma_k Control_{i,t,k}] + \alpha_i + \phi_t + \epsilon_{i,t}$$

  - β significantly $\neq 0$, (relevance).
  - *Clouds* only has indirect effect on deforestation through its effect on LE, *i.e.* $\text{cov}(Clouds_{i,t}, \tilde{\epsilon}_{i,t}) = 0$. (exclusion)
  - May use *Clouds* as instrument.

Empirical strategy

III

- Second-stage regression:

$$DFor_{i,t} = \delta Cloudc_{i,t-1} + \sum_k [\theta_k Control_{i,t,k}] + \psi_i + \lambda_t + \xi_{i,t}$$

- Estimated coefficient $\tilde{\beta}$ of OLS equation (1) not significantly different from 0, suggesting law enforcement does not affect deforestation. Because of reverse causality, expect OLS upward biased.
- Estimated β in first-stage regression significantly negative (see Table 1 of Assunção et al. [2017])

Empirical strategy

IV

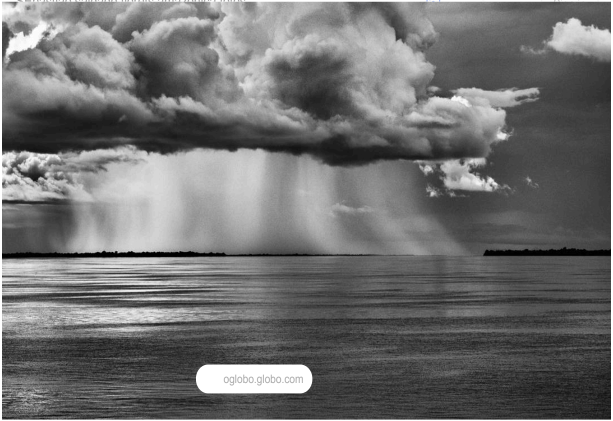
- Elasticity of deforestation with respect to law enforcement (proxied by Cloudc) is .53 for the average municipality.
- Assunção et al. [2017] estimates that DETER reduced deforestation rates by 85% saving 10 billion tons of emissions in a decade.
- Using the full budget of IBAMA, a govt. agency whose duties includes curbing deforestation but also environmental licensing, authorizing use of natural resources, etc..., Assunção et al. [2017] produces a cost estimate of less than \$1/ton.

Empirical strategy

V

- Low estimated cost is a result of the scale economies associated with surveillance using satellite base systems.
- Empirical evidence that environmental law enforcement effectively curbed tropical deforestation in 2006-2016.

Sebastião Salgado



Estimating spatial amplification of degradation of Amazon

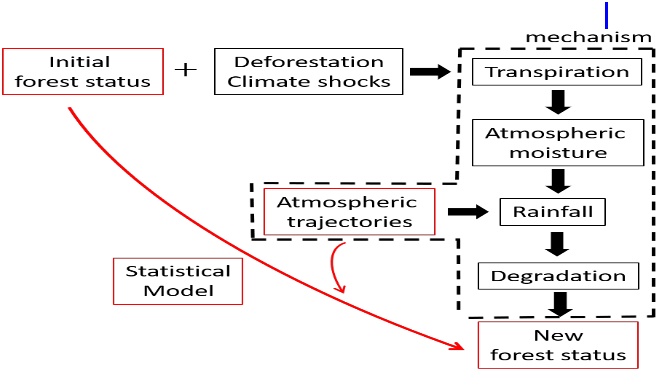
- Araujo et al. [2023]
- Amazon rainforests have been undergoing unprecedented levels of human-induced disturbances.
- In addition to local impacts, such changes are likely to cascade following the eastern-western atmospheric flow generated by trade winds.
  - Trees recycle humidity.
- Model spatial and temporal interactions created by this flow to estimate spread of effects from local disturbances to downwind locations. .

Estimating spatial amplification of degradation of Amazon

II

- Spatial component captures cascading effects propagated by neighboring regions while the temporal component captures the persistence of local disturbances.
- Estimate that on average, the presence of cascading effects mediated by winds in the Amazon doubles the impact of an initial damage.
- Heterogeneity: damage in some regions does not propagate. In others amplification can reach 250%.
- Only account for spillovers mediated by wind ⇒ underestimation of spillovers kil.

Mechanisms and model



- Statistical model abstract from transpiration mechanism; uses variations on atmospheric trajectory to estimate dynamics

Atmospheric trajectories

- Let G be a matrix such that $G_{ij} = 1$ if and only if pixels i and j are immediate neighbors. Otherwise $G_{ij} = 0$
- Let $C_{ij,t} = 1$ if a trajectory of atmospheric circulation at t goes through j before reaching i. Otherwise $C_{ij,t} = 0$.
- Let $W_t = G \circ C_t$, $w_{ij,t} = G_{ij} C_{ij,t} \in \{0, 1\}$ equals 1 if pixel j has **direct** effect on i via atm. circulation at time t. W_t, climate adjacency matrix
- $$\hat{W}_t^{[k]} := \left(\mathbb{I}(G^k > 0) - \sum_{j=1}^{k-1} \mathbb{I}(G^j > 0) \right) \odot C_t$$

Atmospheric trajectories II

- $(W_t^{[k]})_{ij} := (\hat{W}_t^{[k]})_{ij}$, if $i \neq j$ and $(W_t^{[k]})_{ii} := 0$.
- $W_t^{[k]}$ as the matrix of kth-degree neighbors. $W_t^{[k]}$ identifies pixels i and j that are k-th degree geographical neighbors *and* are connected through atmospheric circulation at t.
- $W^{[1]} := W$

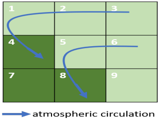
Atmospheric trajectories

III

- To build back-trajectories of wind over time, use monthly three-dimensional wind data (direction and speed) from 1985 to 2013 (Copernicus [2017]). For each pixel of 0.25° resolution in the Amazon ($\sim 27.5km \times 27.5km$ near equator), trace back the trajectory that arrives at that pixel at 800 hPa ($\sim 6000feet$) [Spracklen et al., 2012] for 5 days; enough for back-trajectories to reach ocean.
- Identification assumption: $(W_t^{[k]})_{ij}(W_t^{[k]})_{ji} = 0$, verified in data.

Illustration

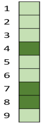
Forest



$$C_t = \begin{array}{c|ccccccccc} & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \hline 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 4 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 5 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 7 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 8 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 9 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}$$

$$G = \begin{array}{c|ccccccccc} & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \hline 1 & 0 & 1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 3 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 4 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 1 & 0 \\ 5 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 6 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 1 & 1 \\ 7 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 \\ 8 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 \\ 9 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 0 \end{array}$$

$Y_t =$



$$W_t = \begin{array}{c|ccccccccc} & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \hline 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 4 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 5 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 7 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 8 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 9 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}$$

$$W_t^{[2]} = \begin{array}{c|ccccccccc} & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \hline 1 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 4 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 5 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 7 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 8 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 9 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}$$

- Y_t vector of pixels LAI at t. C_t shows for each row-pixel, pixels that affect it. $G_{ij} = 1$ if pixels (i, j) are immediate geographical neighbors.

Leaf area index

- LAI= Leaf area/ground area
  - Leaf area is one sided.
- Provides information on density and extent of vegetation cover. Higher LAI value indicates denser canopy with more leaves, which can result in increased photosynthesis, carbon sequestration, biomass production, and overall productivity.
- Higher frequency fluctuations
- Affected by seasons.

Leaf area index II

- LAI data from the NOAA Climate Data Record of AVHRR Leaf Area Index (LAI) at the spatial resolution of 0.05° for 1985-2013 Claverie et al. [2016].
  - Longest LAI time-series available; includes decades of high deforestation rates.
- Reduce LAI's data resolution to wind-data resolution applying an average kernel.

Model

- Y_t vector of LAI indices of pixels.
- Specify the evolution of forest status as a spatial dynamic panel:

$$Y_t = \alpha Y_{t-1} + \sum_{k=1}^{K} \beta_k W_t^{[k]} Y_t + X\gamma + \varepsilon_t \tag{2}$$

- X vector of pixel characteristics
- Consistency requirement for cross-section of LAI, given previous LAI and shocks.
  - 5 days are enough for air parcel to travel entire Amazon

Model

II

- Since dynamic panel use first differences

$$\Delta Y_t = \alpha \Delta Y_{t-1} + \sum_{k=1}^{K} \beta_k \Delta(W_t^{[k]} Y_t) + \tilde{\varepsilon}_t \tag{3}$$

- Use Y_{t-2} as instrument for ΔY_{t-1} because Y_{t-1} is correlated with $\tilde{\varepsilon}_t$
- See Arellano [1989], Phillips and Han [2015]
- To avoid the effects of anthropogenic processes that affect simultaneously several pixels, instrument $\Delta(W_t^{[k]} Y_t)$ by $\Delta W_t^{[k]} Y_0$.

Model III

- Thus use the following $K+1$ moment conditions to identify parameters.

$$\mathbb{E}\left[\tilde{\epsilon}_t Y_{t-2}\right] = 0 \tag{4}$$

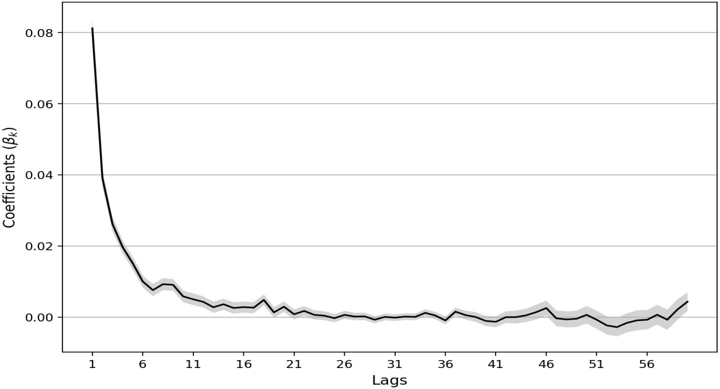
$$\mathbb{E}\left[\tilde{\epsilon}_t \Delta\left(W_t^{[k]} Y_0\right)\right] = 0 \tag{5}$$

- No over-identifying restrictions.

Estimates: α

- Standard errors clustered at the pixel level. Number of observations: 3,300,494. Number of clusters: 9,539
- $\alpha = .22\ (.0019)$
- Persistence coefficient higher than average persistence of individual effects computed by Boulton et al. [2022], but within within 1sd bounds of these individual coefficients.
  - Increase in α = loss in resilience
- α should be interpreted as partial persistence coefficient after controlling for state of rest of forest.

Estimates: β



- $\sum_{k=1}^{20} \beta_k = .26$

Impulse response

- $W^{[k]} := \frac{1}{T}\sum_t W_t^{[k]}$, T time-length of data.
  - Process of atmospheric transport is mean-ergodic
  - Sample average ($W^{[k]}$) representative of future patterns.
- $\Omega := \left(I - \sum_{k=1}^{K} \beta_k W^{[k]}\right)^{-1}$.
  - Summarizes in a single matrix all average feedback effects of the spatial dimension.
- If $||W|| < 1$ and $K = 1$
  $(I - \beta_1 W)^{-1} = I + \beta_1 W + \beta_1^2 W^2 + \beta_1^3 W^3 + ...$
- $$Y_t = \alpha \Omega Y_{t-1} + \Omega X \gamma + \Omega \epsilon_t$$

Impulse response II

- Iterating for a time window Δ,

$$Y_{t+\Delta} = \alpha^{\Delta+1}\Omega^{\Delta+1}Y_{t-1} + \sum_{i=0}^{\Delta}\alpha^{i}\Omega^{i+1}X\gamma + \sum_{i=0}^{\Delta}\alpha^{i}\Omega^{i+1}\epsilon_{t+\Delta-i} \tag{6}$$

- Using previous estimates we can calculate an impulse response function ϕ

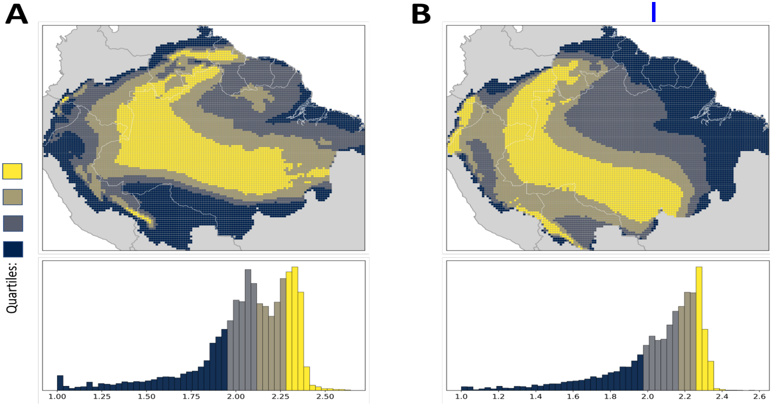
$$\phi(t+\Delta) := \alpha^{\Delta}\mathbb{I}_{row}\left(\Omega^{\Delta+1}\epsilon_t\right) \tag{7}$$

  - $\mathbb{I}_{row}$ denote a row vector of ones

Impulse response III

- $(\phi(\Delta))$ returns the effect that a shock at time t on the forest status over all pixels, for an average pattern of atmospheric situation.
- As $\Delta \to 0$, effect of a unit shock ϵ^i on forest is given by $\sum_j \Omega_{ij}$
  - *Index of influence* of i
- $\sum_i \Omega_{ij}$
  - *Index of exposure* of j

A: Index of influence B: Index of exposure

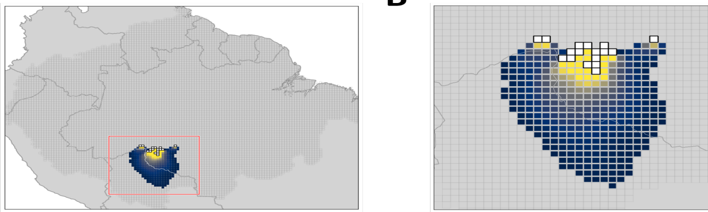


- Average multiplier is 2.

Transnational externality

- Effect of deforestation in pixels in Rondonia state.

A B



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IV

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